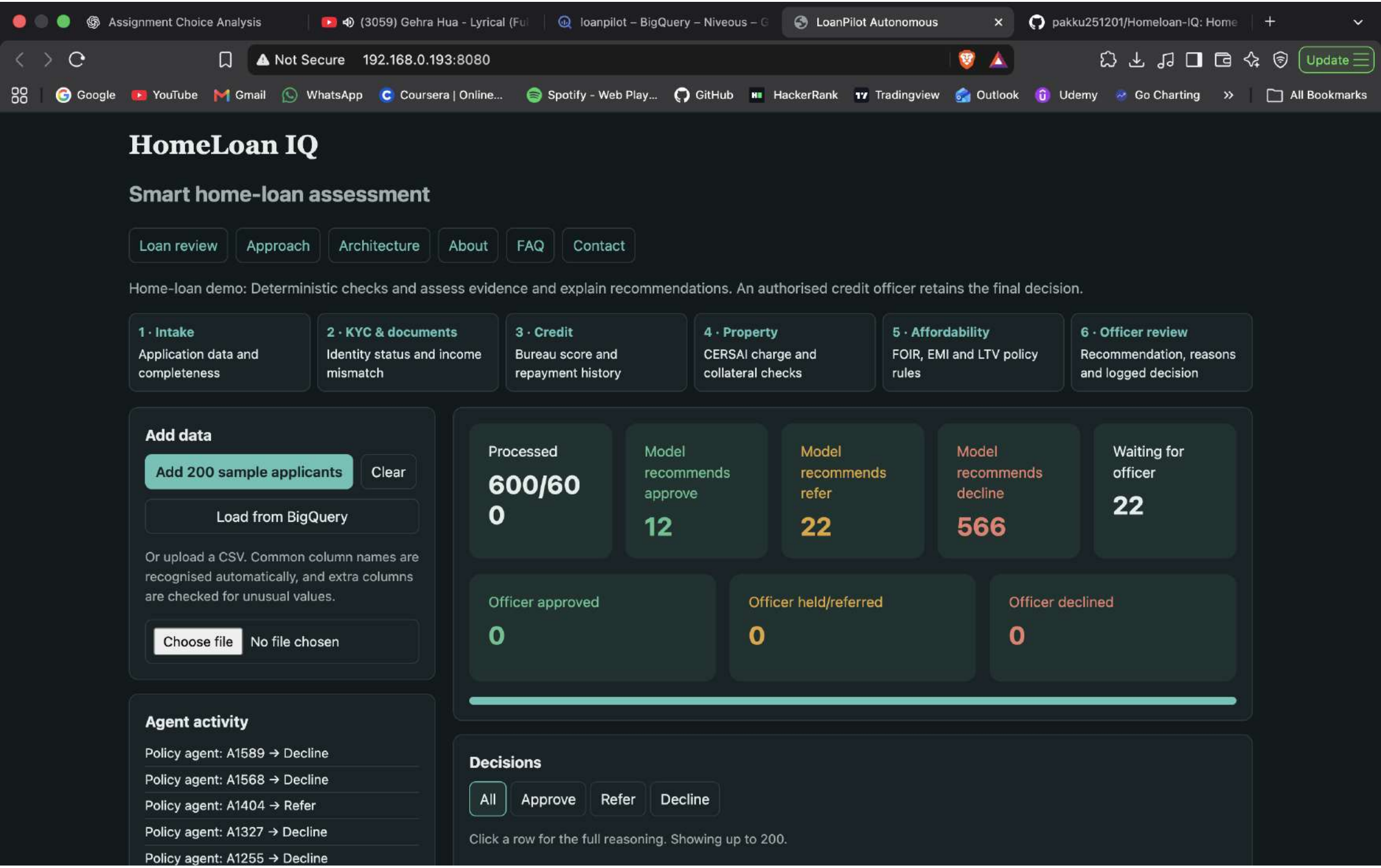

Assignment Choice Analysis
(3059) Gehra Hua - Lyrical (Ful
loanpilot – BigQuery – Niveous – G
LoanPilot Autonomous
pakku251201/Homeloan-IQ: Home
Not Secure
192.168.0.193:8080
Update
Google
YouTube
Gmail
WhatsApp
Coursera | Online...
Spotify - Web Play...
GitHub
HackerRank
Tradingview
Outlook
Udemy
Go Charting
All Bookmarks
HomeLoan IQ
Smart home-loan assessment
Loan review
Approach
Architecture
About
FAQ
Contact
Home-loan demo: Deterministic checks and assess evidence and explain recommendations. An authorised credit officer retains the final decision.
1 · Intake
Application data and completeness
2 · KYC & documents
Identity status and income mismatch
3 · Credit
Bureau score and repayment history
4 · Property
CERSAI charge and collateral checks
5 · Affordability
FOIR, EMI and LTV policy rules
6 · Officer review
Recommendation, reasons and logged decision
Add data
Add 200 sample applicants
Clear
Load from BigQuery
Or upload a CSV. Common column names are recognised automatically, and extra columns are checked for unusual values.
Choose file
No file chosen
Processed
600/600
Model recommends approve
12
Model recommends refer
22
Model recommends decline
566
Waiting for officer
22
Officer approved
0
Officer held/referred
0
Officer declined
0
Agent activity
Policy agent: A1589 → Decline
Policy agent: A1568 → Decline
Policy agent: A1404 → Refer
Policy agent: A1327 → Decline
Policy agent: A1255 → Decline
Decisions
All
Approve
Refer
Decline
Click a row for the full reasoning. Showing up to 200.

Assignment Choice Analysis | (3059) Gehra Hua - Lyrical (Ful | loanpilot – BigQuery – Niveous – G | LoanPilot Autonomous | pakku251201/Homeloan-IQ: Home
Not Secure 192.168.0.193:8080
Update
Google YouTube Gmail WhatsApp Coursera | Online... Spotify - Web Play... GitHub HackerRank Tradingview Outlook Udemy Go Charting All Bookmarks

Income Verification Section 6.3
Income documents must match bank credits. Any mismatch goes to an officer.

Credit Policy Section 3.4
Three or more bounced payments in 12 months go to an officer.

KYC Master Direction
PAN and Aadhaar must be verified before any sanction.

Data Quality Standard Section 1
Missing, impossible or unusual data is never decided automatically. It goes to an officer.

Case A1382: Refer

No hard failure, but an officer must resolve the flags.

Data provenance

Logical sources used by this demo. Bureau and CERSAI are simulated. Confirm BigQuery is loaded before treating this as a live source trace.

Application / customer	Account / repayments	Bureau (mock)	CERSAI (mock)
Applicant ID, name, income, requested amount, property value, tenure	Existing EMI and bounced-payment count	Credit score; synthetic bureau data	Existing property charge indicator

Applicant CSV / BigQuery view
Values loaded for this case; uploaded CSVs use mapped column names

Workflow trace

Intake agent: all key fields present; impossible: loan is larger than property value.
Recommendation: Refer. Reasons below.

Check	Result	Why	Policy used
Data needs a human check	Flag	loan is larger than property value	**Data Quality Standard Section 1** Missing, impossible or unusual data is never decided automatically. It goes to an officer.

Credit officer decision: Approve | Hold | Decline

Assignment Choice Analysis | (3059) Gehra Hua - Lyrical (Ful | loanpilot – BigQuery – Niveous – G | LoanPilot Autonomous | pakku251201/Homeloan-IQ: Home
Not Secure 192.168.0.193:8080
Update
Google YouTube Gmail WhatsApp Coursera | Online... Spotify - Web Play... GitHub HackerRank Tradingview Outlook Udemy Go Charting All Bookmarks

goes to legal review before sanction.

Income Verification Section 6.3
Income documents must match bank credits. Any mismatch goes to an officer.

Credit Policy Section 3.4
Three or more bounced payments in 12 months go to an officer.

KYC Master Direction
PAN and Aadhaar must be verified before any sanction.

Data Quality Standard Section 1
Missing, impossible or unusual data is never decided automatically. It goes to an officer.

Case A1065: Approve

All checks passed. Suggest sanction at ₹40,50,000, EMI ₹33,297.

Data provenance

Logical sources used by this demo. Bureau and CERSAI are simulated. Confirm BigQuery is loaded before treating this as a live source trace.

Application / customer
Applicant ID, name, income, requested amount, property value, tenure

Account / repayments
Existing EMI and bounced-payment count

Bureau (mock)
Credit score; synthetic bureau data

CERSAI (mock)
Existing property charge indicator

Applicant CSV / BigQuery view
Values loaded for this case; uploaded CSVs use mapped column names

Workflow trace

Intake agent: all key fields present.
Document check: KYC verified, documents consistent.
Data agent: bureau score 787, 0 bounces, CERSAI clear (mock services).
Credit and policy agent: applied P1, P2, P3 and checked flags.
Recommendation: Approve. Ready for officer sign-off.

Check	Result	Why	Policy used
Credit score	Pass	Score 787 against minimum 700	**Credit Policy Section 3.1** Bureau (CIBIL) score must be 700 or above.
Repayment burden (FOIR)	Pass	48% of income against limit 50%. New EMI ₹33,297	**Credit Policy Section 4.2** Total monthly EMIs after the new loan must not exceed 50% of monthly income (FOIR).
Loan-to-value	Pass	70% of property value against limit 80%	**RBI LTV norms** Loan up to ₹30 lakh: LTV up to 90%. ₹30 to 75 lakh: up to 80%. Above ₹75 lakh: up to 75%.

Credit officer decision: Approve Hold Decline

Assignment Choice Analysis | (3059) Gehra Hua - Lyrical (Ful | loanpilot – BigQuery – Niveous – G | LoanPilot Autonomous | pakku251201/Homeloan-IQ: Home
Not Secure 192.168.0.193:8080
Update
Google YouTube Gmail WhatsApp Coursera | Online... Spotify - Web Play... GitHub HackerRank Tradingview Outlook Udemy Go Charting All Bookmarks

Income Verification Section 6.3
Income documents must match bank credits. Any mismatch goes to an officer.

Credit Policy Section 3.4
Three or more bounced payments in 12 months go to an officer.

KYC Master Direction
PAN and Aadhaar must be verified before any sanction.

Data Quality Standard Section 1
Missing, impossible or unusual data is never decided automatically. It goes to an officer.

Case A1006: Decline

Does not meet policy. Counter-offer: ₹8,42,079, the largest loan that keeps repayments within 50% of income.

Data provenance

Logical sources used by this demo. Bureau and CERSAI are simulated. Confirm BigQuery is loaded before treating this as a live source trace.

Application / customer
Applicant ID, name, income, requested amount, property value, tenure

Account / repayments
Existing EMI and bounced-payment count

Bureau (mock)
Credit score; synthetic bureau data

CERSAI (mock)
Existing property charge indicator

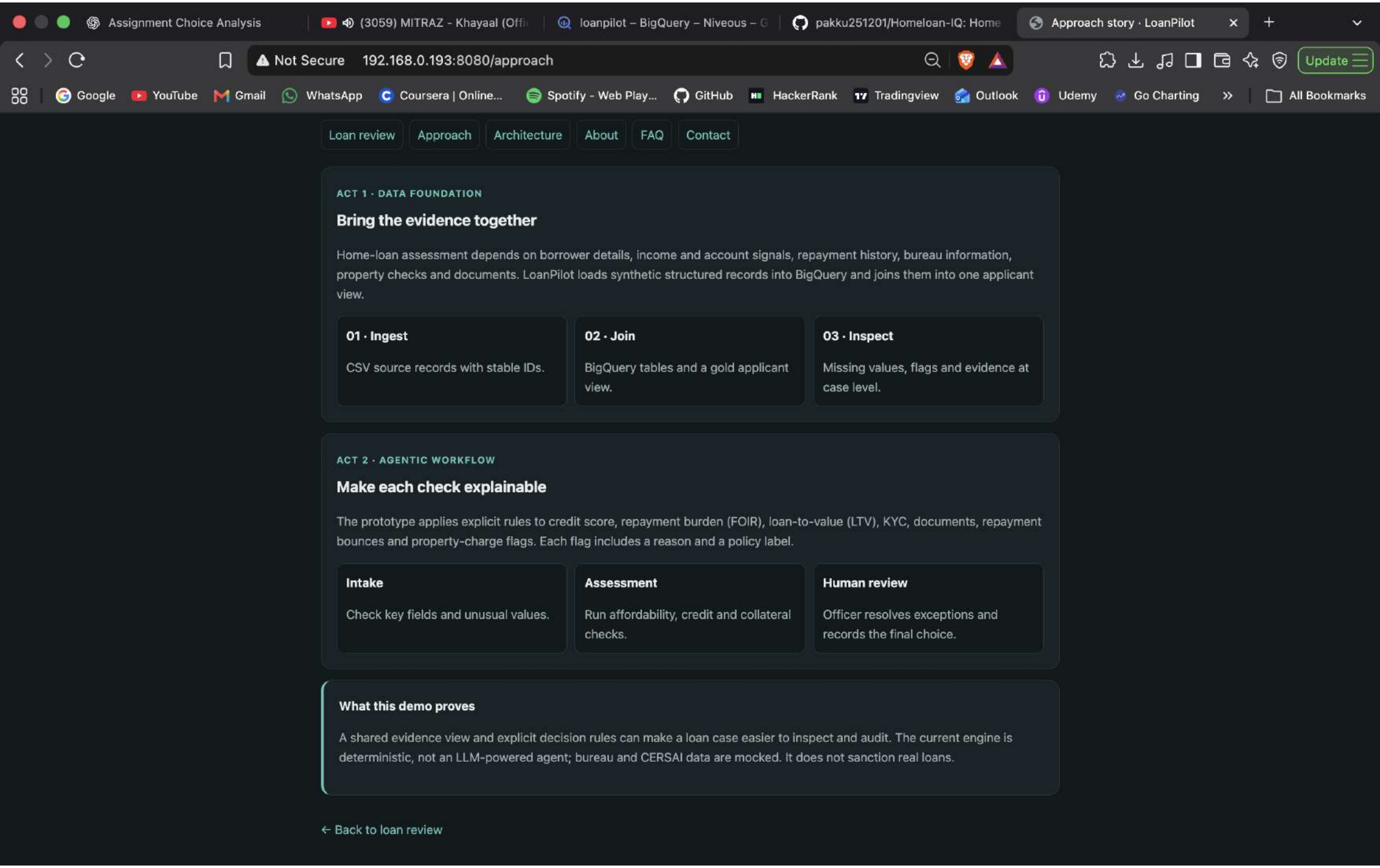
Applicant CSV / BigQuery view
Values loaded for this case; uploaded CSVs use mapped column names

Workflow trace

Intake agent: all key fields present.
Document check: KYC verified, documents consistent.
Data agent: bureau score 756, 0 bounces, CERSAI clear (mock services).
Credit and policy agent: applied P1, P2, P3 and checked flags.
Recommendation: Decline. Reasons below.

Check	Result	Why	Policy used
Credit score	Pass	Score 756 against minimum 700	**Credit Policy Section 3.1** Bureau (CIBIL) score must be 700 or above.
Repayment burden (FOIR)	Fail	79% of income against limit 50%. New EMI ₹19,426	**Credit Policy Section 4.2** Total monthly EMIs after the new loan must not exceed 50% of monthly income (FOIR).
Loan-to-value	Pass	79% of property value against limit 90%	**RBI LTV norms** Loan up to ₹30 lakh: LTV up to 90%. ₹30 to 75 lakh: up to 80%. Above ₹75 lakh: up to 75%.

Credit officer decision: Approve Hold Decline


Assignment Choice Analysis
(3059) MITRAZ - Khayaal (Offi
loanpilot – BigQuery – Niveous – G
pakku251201/Homeloan-IQ: Home
Approach story · LoanPilot
Not Secure
192.168.0.193:8080/approach
Update
Google
YouTube
Gmail
WhatsApp
Coursera | Online...
Spotify - Web Play...
GitHub
HackerRank
Tradingview
Outlook
Udemy
Go Charting
All Bookmarks
Loan review
Approach
Architecture
About
FAQ
Contact
ACT 1 · DATA FOUNDATION
Bring the evidence together
Home-loan assessment depends on borrower details, income and account signals, repayment history, bureau information, property checks and documents. LoanPilot loads synthetic structured records into BigQuery and joins them into one applicant view.
01 · Ingest
CSV source records with stable IDs.
02 · Join
BigQuery tables and a gold applicant view.
03 · Inspect
Missing values, flags and evidence at case level.
ACT 2 · AGENTIC WORKFLOW
Make each check explainable
The prototype applies explicit rules to credit score, repayment burden (FOIR), loan-to-value (LTV), KYC, documents, repayment bounces and property-charge flags. Each flag includes a reason and a policy label.
Intake
Check key fields and unusual values.
Assessment
Run affordability, credit and collateral checks.
Human review
Officer resolves exceptions and records the final choice.
What this demo proves
A shared evidence view and explicit decision rules can make a loan case easier to inspect and audit. The current engine is deterministic, not an LLM-powered agent; bureau and CERSAI data are mocked. It does not sanction real loans.
← Back to loan review

Architecture

A simple, auditable path from sources to officer action

Loan review | Approach | Architecture | About | FAQ | Contact

Logical architecture

Sources

Applications · Customers · Accounts · Repayments · Mock bureau · Mock CERSAI

↓ CSV ingestion

Data platform · BigQuery

Six source tables → SQL joins/aggregation → `applicant_view`

↓ Flask API

Loan review UI

Applicant selection · evidence · rule outcomes · explanations · activity log

↓ officer decision

Audit record

`decision_log` stores application ID, officer choice and timestamp

Controls and boundaries

- Only synthetic applicant data and mock external responses.
- Explicit, deterministic checks; reasons are displayed alongside policy labels.
- Officer decision is separate from the system recommendation.
- BigQuery integration is optional for local fallback, but the live data path must be verified before presenting it as working.
- No OCR, vector retrieval, live bureau/CERSAI APIs, authentication or production security in this prototype.



Controls and boundaries

- Only synthetic applicant data and mock external responses.
- Explicit, deterministic checks; reasons are displayed alongside policy labels.
- Officer decision is separate from the system recommendation.
- BigQuery integration is optional for local fallback, but the live data path must be verified before presenting it as working.
- No OCR, vector retrieval, live bureau/CERSAI APIs, authentication or production security in this prototype.

Next production controls

Identity and role-based access, encryption, versioned policy sources, immutable audit events, fairness testing, model validation, monitoring, secure document handling and formal compliance review.

About LoanPilot

A proof of concept for home-loan operations

Purpose

LoanPilot illustrates how structured lending data and explicit policy checks can support an officer through early home-loan assessment. It is designed for a presales discussion, not for real lending.

What is included

- 600 generated applicants in the source-data generator.
- Mock bureau and property-charge checks.
- BigQuery joined applicant view and officer-decision log.
- Case-level reasons and a manual officer decision control.

Responsible use

All records are synthetic. Recommendations are demonstration outputs, not credit advice, actual eligibility determinations or regulatory interpretations.

← Back to loan review

Frequently asked questions

Short answers about the demo

Is this a real AI model?

No. The current checks are deterministic JavaScript rules. "Agent" labels describe workflow roles, not a deployed LLM.

Does it connect to CIBIL or CERSAI?

No. These are synthetic/mock records. There are no live external-service calls.

Does BigQuery store the data?

The loader is designed to create six source tables, a joined applicant view and a decision log in the configured Google Cloud project. Verify table/view counts and the app's Load from BigQuery button before claiming live integration.

Can the system approve a loan?

No final sanction should be made by this demo. It gives a rule-based recommendation; an authorised officer makes the final decision.

Are the policy thresholds official?

Some UI thresholds are illustrative. Check the cited official RBI material and the relevant lender's current policy before treating any threshold as authoritative.

Why does it sometimes generate sample applicants?

The page falls back to synthetic sample records if the BigQuery API cannot return data, so the UI can still be demonstrated offline.

Contact and feedback

Pratheek P Rao / pratheekprao@gmail.com

For walkthrough feedback

This is an individual assignment proof of concept. For a submission, provide your own name and contact details in the README or presentation rather than exposing personal details in a public demo by default.

Suggested feedback: usability of the loan journey, clarity of the evidence and policy reasons, missing checks, and points requiring human review.

Security note

Do not upload real customer documents or personal financial data. This prototype is not designed for confidential production use.

← Back to loan review